Homework week 6, core econometrics, fall 2025

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1. **Two Stage Least Squares**:
   Consider the model

   $$Y = \beta_1 X_1 + \beta_2 X_2 + U,$$

   in which it is known that $\mathrm{E}(U) = 0$. We also know that $\mathrm{E}(Z_1 U) = 0$ and $\mathrm{E}(Z_2 U) = 0$. You may assume that the observed variables have mean zero, and the data on (Y, X_1, X_2, Z_1, Z_2) are independent and identically distributed over $i = 1, ..., n$.

   (a) Briefly explain how to calculate the two stage least squares (2SLS) estimator of the parameter vector $\beta = (\beta_1, \beta_2)'$.

   (b) With reference to the first stage equation for X_1, state a necessary condition for the 2SLS estimator to estimate the parameter vector β consistently in this model.

   (c) With reference to the first stage equation for X_2, state a necessary condition for the 2SLS estimator to estimate the parameter vector β consistently in this model.

   (d) State a necessary and sufficient condition for the 2SLS estimator to estimate the parameter vector β consistently in this model, as a condition on the rank of a 2×2 matrix.

2. **Control Function**:
Consider the linear model with a single endogenous explanatory variable (X, with $E(XU) \neq 0$) and a set of L valid instruments ($Z = (Z_1, Z_2, ..., Z_l)'$, with $E(ZU) = 0$)

$$Y = X\beta + U$$

Consider the linear projections

$$X = Z'\pi + R$$

$$U = R\gamma + V^*$$

Assume that $E(Y) = E(X) = E(Z_l) = 0$ for $l = 1, 2, ..., L$, and that the set of instruments is informative, so that we have $\pi \neq 0$.

(a) Show that $E(XV^*) = E(Z'V^*)\pi$.

(b) Show that $E(Z'V^*) = 0$.

(c) Assuming that π is known, what do these results suggest about the OLS estimator of $(\beta, \gamma)'$ in the augmented model

$$Y = X\beta + R\gamma + V^* \text{ ?}$$

(d) What does this suggest about the OLS estimator of $(\beta, \gamma)'$ in the augmented model in which R is replaced by $\hat{R}$, the residuals from the first stage linear projection model estimated using OLS?